

IP Addressing (Part 1)

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How the Internet Works - IP Addressing

- Because the Internet is a global network of computers each computer connected to the Internet **must** have a unique address. **Internet addresses are in the form nnn.nnn. nnn. nnn where nnn must be a number from 0 - 255. This address is known as an IP address.**
- Each computer on the Internet has a unique identification number, called an IP (Internet Protocol) address.
- The IP addressing system currently in use on the Internet uses a four-part number.
- For example, 106.29.242.17

IP Address

- An IPv4 address is a **32-bit address** that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.
- Two devices on the Internet can never have the same address at the same time.
- IP addressing identifies hosts and network devices.
- IPv4 uses 32-bit addresses, which means that the **address space is 2^{32}** or 4,294,967,296 (more than 4 billion).

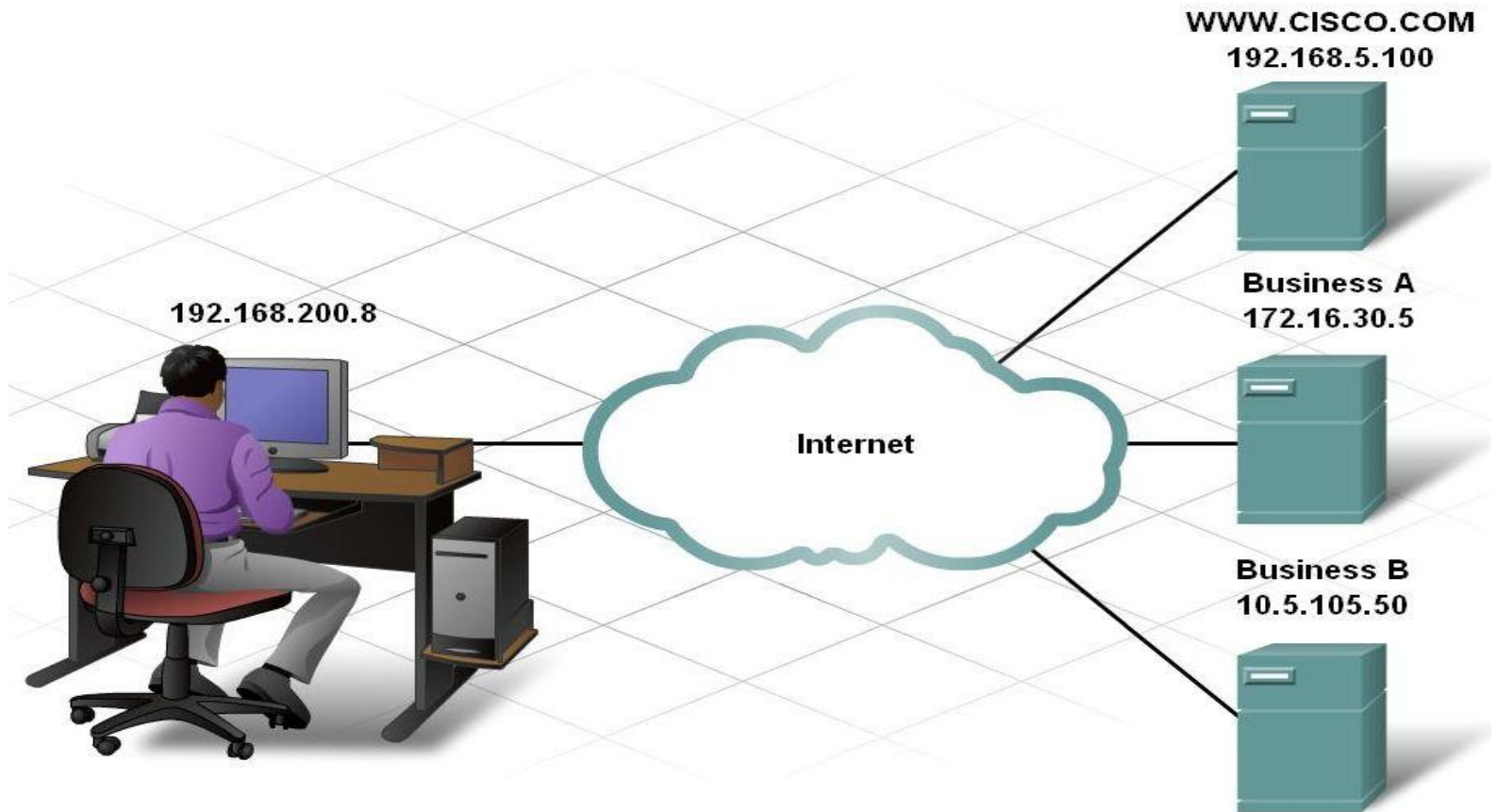
IP Addressing

- The combination of the four IP address parts provides **4.2 billion possible addresses (256 x 256 x 256 x 256)**.
- This number seemed adequate until 1998.
- Members of various Internet task forces are working to develop an alternate addressing system that will accommodate the projected growth.
- However, all of their working solutions require extensive hardware and software changes throughout the Internet.

Requirement of IP

- If you connect to the Internet through an Internet Service Provider (ISP), you are usually **assigned a temporary IP address for the duration of your dial-in session.**
- If you connect to the Internet from a local area network (LAN) your computer might have a permanent IP address or it might obtain a temporary one from a DHCP (Dynamic Host Configuration Protocol) server.
- In any case, if you are connected to the Internet, your computer has a unique IP address.

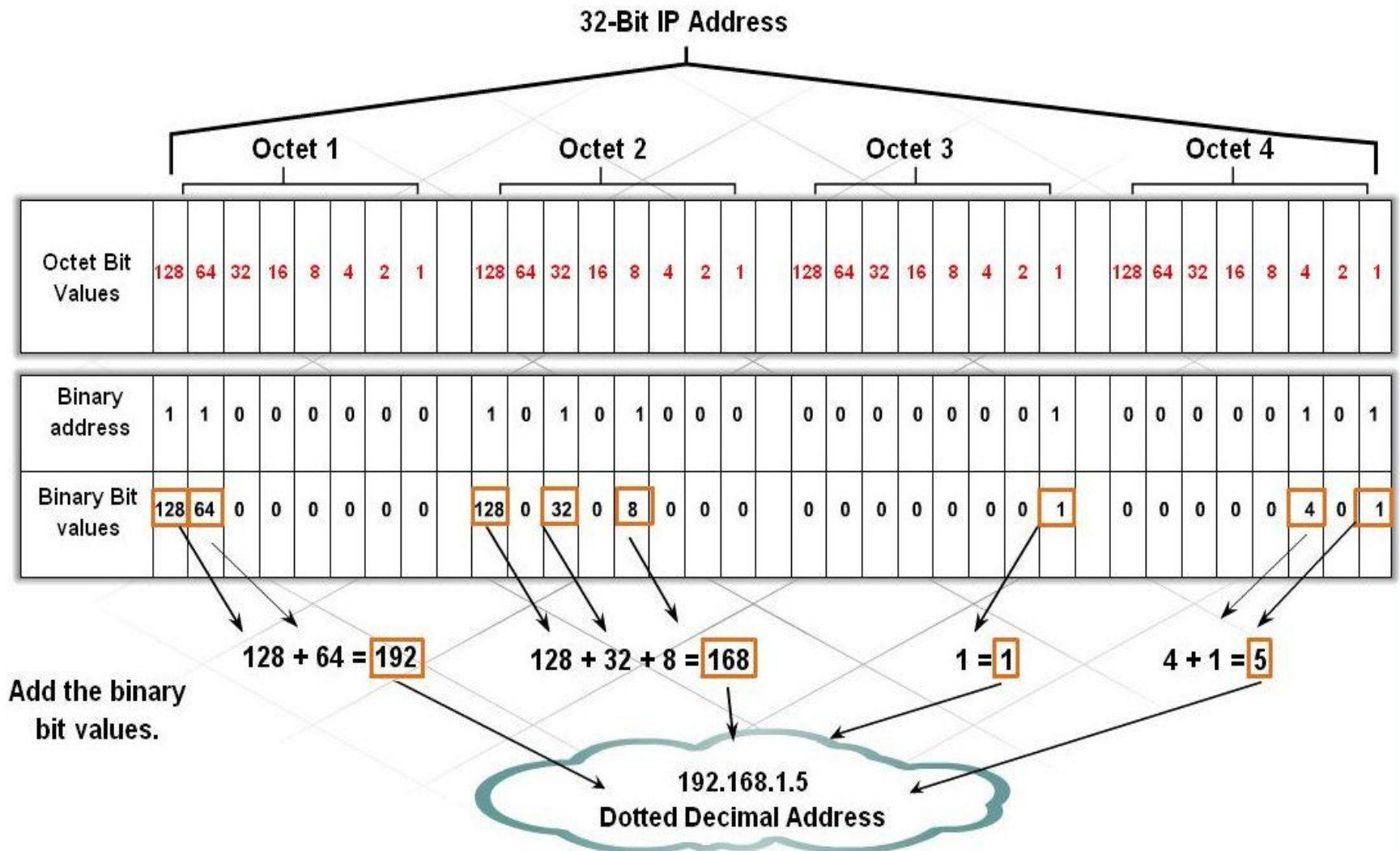
Purpose of IP Addresses



Check IP using- The 'Ping' Program

- If you're using Microsoft Windows or a flavor of Unix and have a connection to the Internet, there is a handy program to see if a computer on the Internet is alive. It's called **ping**.
- If you are using Windows, start a command prompt window. If you're using a flavor of Unix, get to a command prompt. Type **ping www.yahoo.com**.
- The ping program will send a 'ping' (actually an ICMP (Internet Control Message Protocol) echo request message) to the named computer. The pinged computer will respond with a reply.
- The ping program will count the time expired until the reply comes back (if it does). Ping will resolve the domain name and display the computer's IP address..

Structure of IP Address



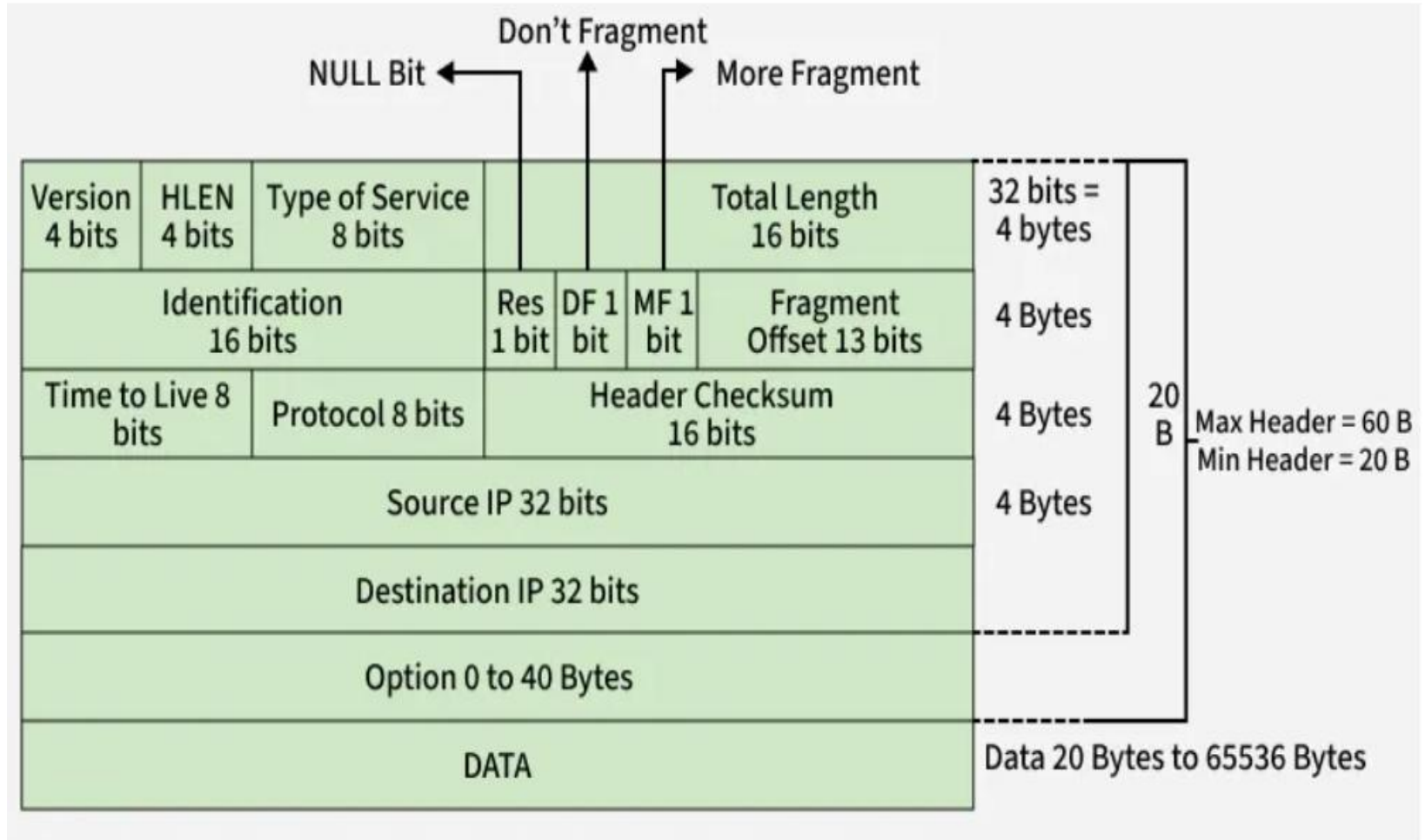
Internet Protocol (IP)

- The Internet Protocol (IP) is the transmission mechanism used by the TCP/IP protocols at the network layer.
- IP is an unreliable and connectionless datagram protocol-a **best-effort delivery** service.
- *best-effort* means that IP packets can be corrupted, lost, arrive out of order, or delayed and may create congestion for the network.
- If reliability is important, IP must be paired with a reliable protocol such as TCP

Datagrams

- Packets in the network (internet) layer are called *datagrams*.
- A datagram is a variable-length packet consisting of two parts: header and data.
- The header is 20 to 60 bytes in length and contains information essential to routing and delivery.

IPv4 Datagram Header



IPv4 Header

Field	Size	Purpose
Version	4 bits	Specifies IP version (IPv4 = 4)
HL (Header Length)	4 bits	Length of header in 32-bit words. The minimum value for this field is 5 and the maximum is 15.
Type of Service (ToS)	8 bits	Defines priority and quality of service
Total Length	16 bits	Total size of IP packet (header + data)
Identification	16 bits	Identifies fragments of the same packet
Flags	3 bits	Controls fragmentation (DF, MF bits)
Fragment Offset	13 bits	Position of fragment in original packet
TTL (Time To Live)	8 bits	Limits packet lifetime (prevents looping)
Protocol	8 bits	Specifies upper layer protocol (TCP=6, UDP=17, etc.)
Header Checksum	16 bits	Error checking of header only
Source Address	32 bits	Sender's IP address
Destination Address	32 bits	Receiver's IP address
Options	Variable	Optional fields (security, timestamp, etc.)
Padding	Variable	Added to make header multiple of 32 bits

IPv4

- Minimum header size = **20 bytes**
- Maximum header size = **60 bytes**
- TTL prevents infinite routing loops
- Checksum checks header only (not data)

Notations: Binary Notation

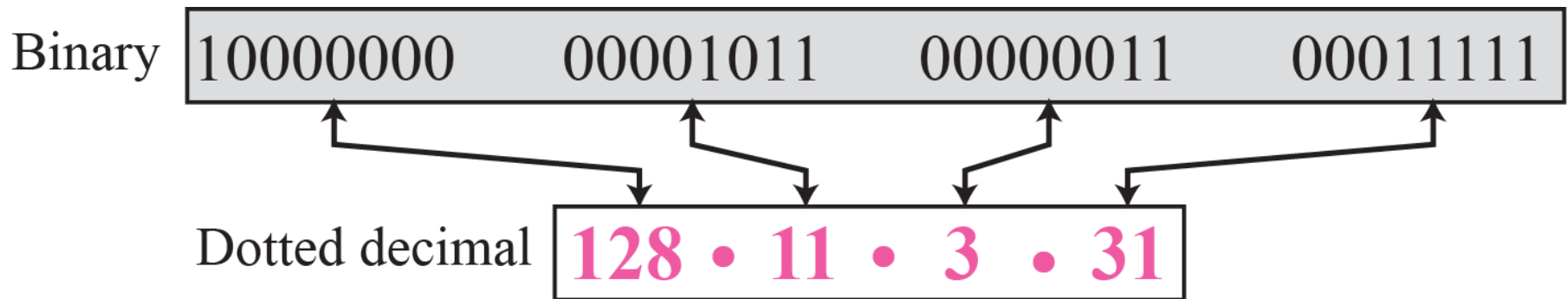
- In binary notation, the IPv4 address is displayed as 32 bits. Each octet is often referred to as a byte. So it is common to hear an IPv4 address referred to as a 32-bit address or a 4-byte address. The following is an example of an IPv4 address in binary notation:

01110101 10010101 00011101 00000010

Dotted-Decimal Notation

- To make the IPv4 address more compact and easier to read, Internet addresses are usually written in decimal form with a decimal point (dot) separating the bytes. The following is the dotted-decimal notation of the above address:

117.149.29.2



Example

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

- a. 10000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 11100111 11011011 10001011 01101111
- d. 11111001 10011011 11111011 00001111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation:

- a. 129.11.11.239
- b. 193.131.27.255
- c. 231.219.139.111
- d. 249.155.251.15

Example

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

c. *241.8.56.12*

d. *75.45.34.78*

Solution

We replace each decimal number with its binary equivalent:

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

c. 11110001 00001000 00111000 00001100

d. 01001011 00101101 00100010 01001110

Example

Find the error, if any, in the following IPv4 addresses:

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution

- a. There should be no leading zeroes (045).
- b. We may not have more than 4 bytes in an IPv4 address.
- c. Each byte should be less than or equal to 255.
- d. A mixture of binary notation and dotted-decimal notation.

Example

Change the following IPv4 addresses from binary notation to hexadecimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 4 bits with its hexadecimal equivalent. Note that 0X (or 0x) is added at the beginning or the subscript 16 at the end.

a. 0X810B0BEF or $810B0BEF_{16}$

b. 0XC1831BFF or $C1831BFF_{16}$

Example

Find the number of addresses in a range if the first address is 146.102.29.0 and the last address is 146.102.32.255.

Solution

We can subtract the first address from the last address in base 256 (see Appendix B). The result is 0.0.3.255 in this base. To find the number of addresses in the range (in decimal), we convert this number to base 10 and add 1 to the result..

$$\text{Number of addresses} = (0 \times 256^3 + 0 \times 256^2 + 3 \times 256^1 + 255 \times 256^0) + 1 = 1024$$

Example

The first address in a range of addresses is 14.11.45.96. If the number of addresses in the range is 32, what is the last address?

Solution

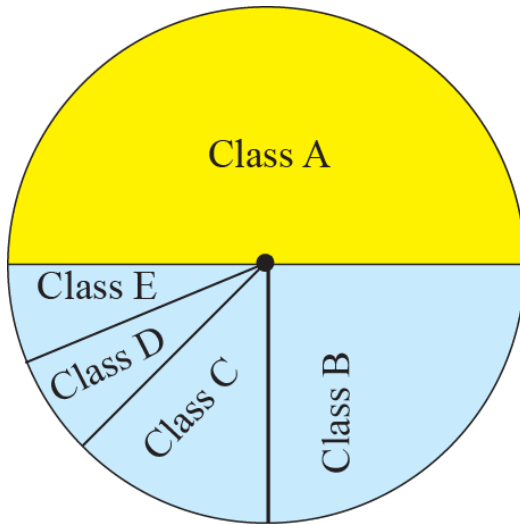
We convert the number of addresses minus 1 to base 256, which is 0.0.0.31. We then add it to the first address to get the last address. Addition is in base 256.

$$\text{Last address} = (14.11.45.96 + 0.0.0.31)_{256} = 14.11.45.127$$

CLASSFUL ADDRESSING

- IP addresses, when started a few decades ago, used the concept of classes.
- This architecture is called classful addressing.
- In the mid-1990s, a new architecture, called classless addressing, was introduced that supersedes the original architecture.
- In this section, we introduce classful addressing because it paves the way for understanding classless addressing and justifies the rationale for moving to the new architecture.

Classful Addressing



Class A: $2^{31} = 2,147,483,648$ addresses, 50%

Class B: $2^{30} = 1,073,741,824$ addresses, 25%

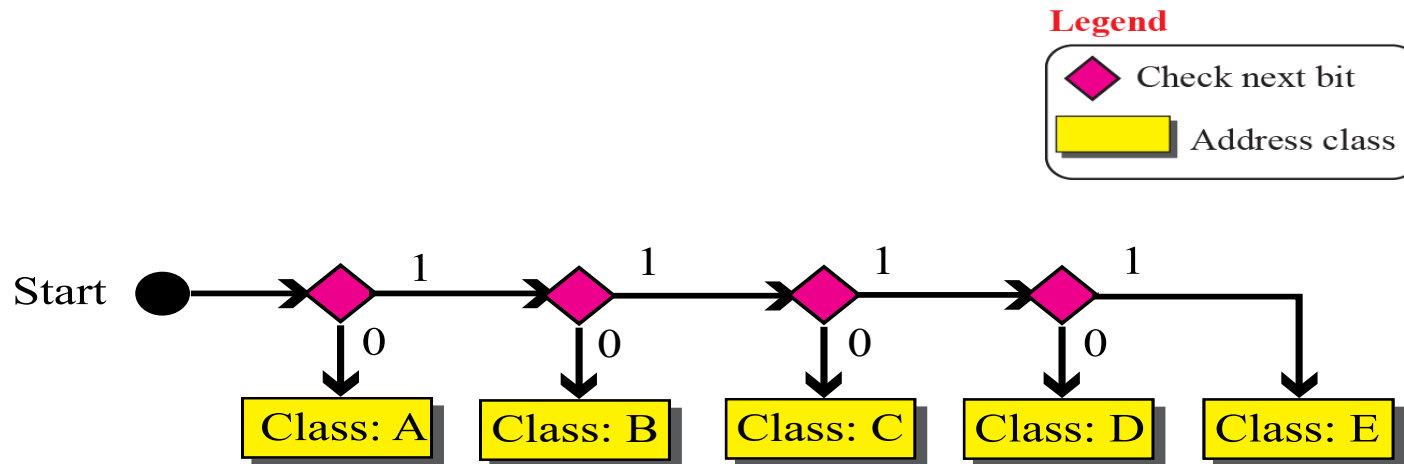
Class C: $2^{29} = 536,870,912$ addresses, 12.5%

Class D: $2^{28} = 268,435,456$ addresses, 6.25%

Class E: $2^{28} = 268,435,456$ addresses, 6.25%

Finding the class of address

	Octet 1	Octet 2	Octet 3	Octet 4		Byte 1	Byte 2	Byte 3	Byte 4
Class A	0.....				Class A	0–127			
Class B	10.....				Class B	128–191			
Class C	110.....				Class C	192–223			
Class D	1110....				Class D	224–255			
Class E	1111....				Class E	240–255			
Binary notation					Dotted-decimal notation				



IPv4 Address Classes Distribution

Class	Starting Bits	Address Range	Total Addresses	% of Total	Use
A	0	0.0.0.0 – 127.255.255.255	$2^{31} = 2,147,483,648$	50%	Very large networks
B	10	128.0.0.0 – 191.255.255.255	$2^{30} = 1,073,741,824$	25%	Medium networks
C	110	192.0.0.0 – 223.255.255.255	$2^{29} = 536,870,912$	12.5%	Small networks
D	1110	224.0.0.0 – 239.255.255.255	$2^{28} = 268,435,456$	6.25%	Multicast
E	1111	240.0.0.0 – 255.255.255.255	$2^{28} = 268,435,456$	6.25%	Experimental

Types of IP Addresses

IP Address Classes					
Address Class	1st octet range (decimal)	1st octet bits (green bits don't change)	Network (N) and Host (H) parts of an address	Default subnet mask (decimal and binary)	Numbers of possible networks and hosts per network
A	1 - 127	00000000 - 01111111	N.H.H.H	255.0.0.0 11111111.00000000.00000000.00000000	126 nets (2^7-2) 16,777,214 hosts per net ($2^{24}-2$)
B	128 - 191	10000000 - 10111111	N.N.H.H	255.255.0.0 11111111.11111111.00000000.00000000	16,382 nets ($2^{14}-2$) 65,534 hosts per net ($2^{16}-2$)
C	192 - 223	11000000 - 11011111	N.N.N.H	255.255.255.0 11111111.11111111.11111111.00000000	2,097,150 nets ($2^{21}-2$) 254 hosts per net (2^8-2)
D	224 - 239	11100000 - 11101111	Not for commercial use as a host		
E	240 - 255	11110000 - 11111111	Not for commercial use as a host		

^^ All zeros (0) and all ones (1) are invalid host addresses.

Example

Find the class of each address:

- a. 00000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 10100111 11011011 10001011 01101111
- d. 11110011 10011011 11111011 00001111

Solution

- a. The first bit is 0. This is a class A address.
- b. The first 2 bits are 1; the third bit is 0. This is a class C address.
- c. The first bit is 1; the second bit is 0. This is a class B address.
- d. The first 4 bits are 1s. This is a class E address.

Example

Find the class of each address:

a. 227.12.14.87

b. 193.14.56.22

c. 14.23.120.8

d. 252.5.15.111

Solution

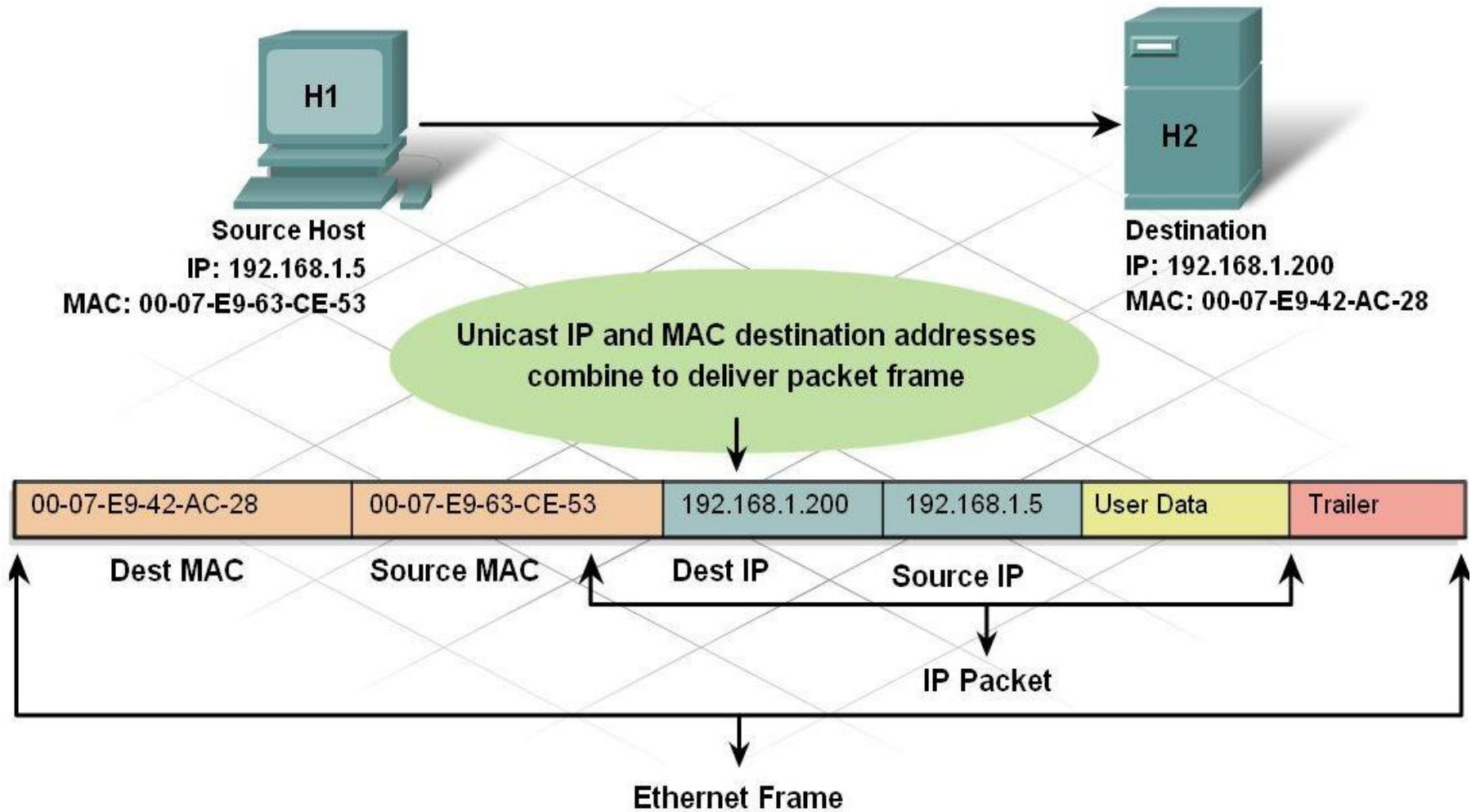
a. The first byte is 227 (between 224 and 239); the class is D.

b. The first byte is 193 (between 192 and 223); the class is C.

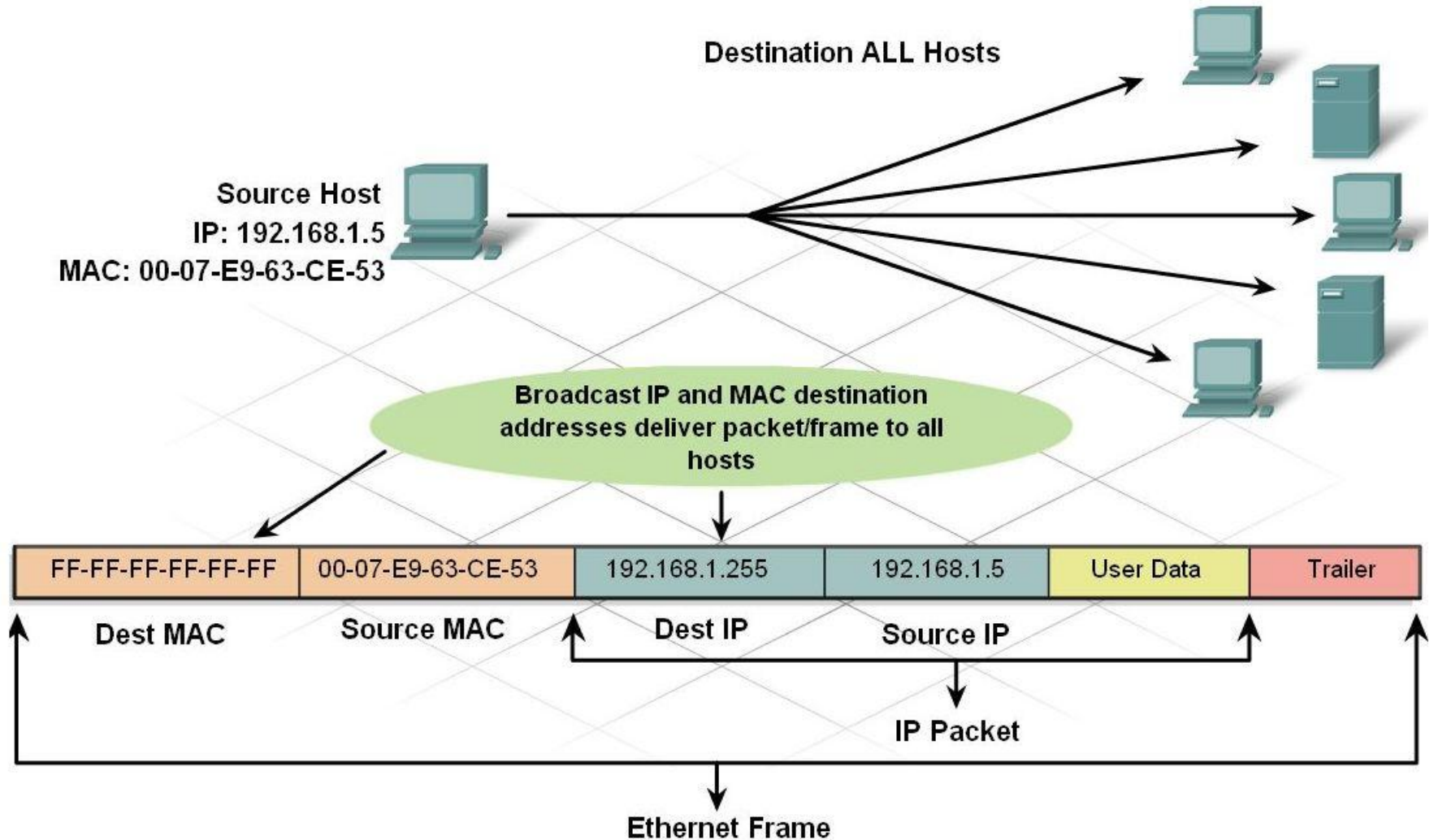
c. The first byte is 14 (between 0 and 127); the class is A.

d. The first byte is 252 (between 240 and 255); the class is E.

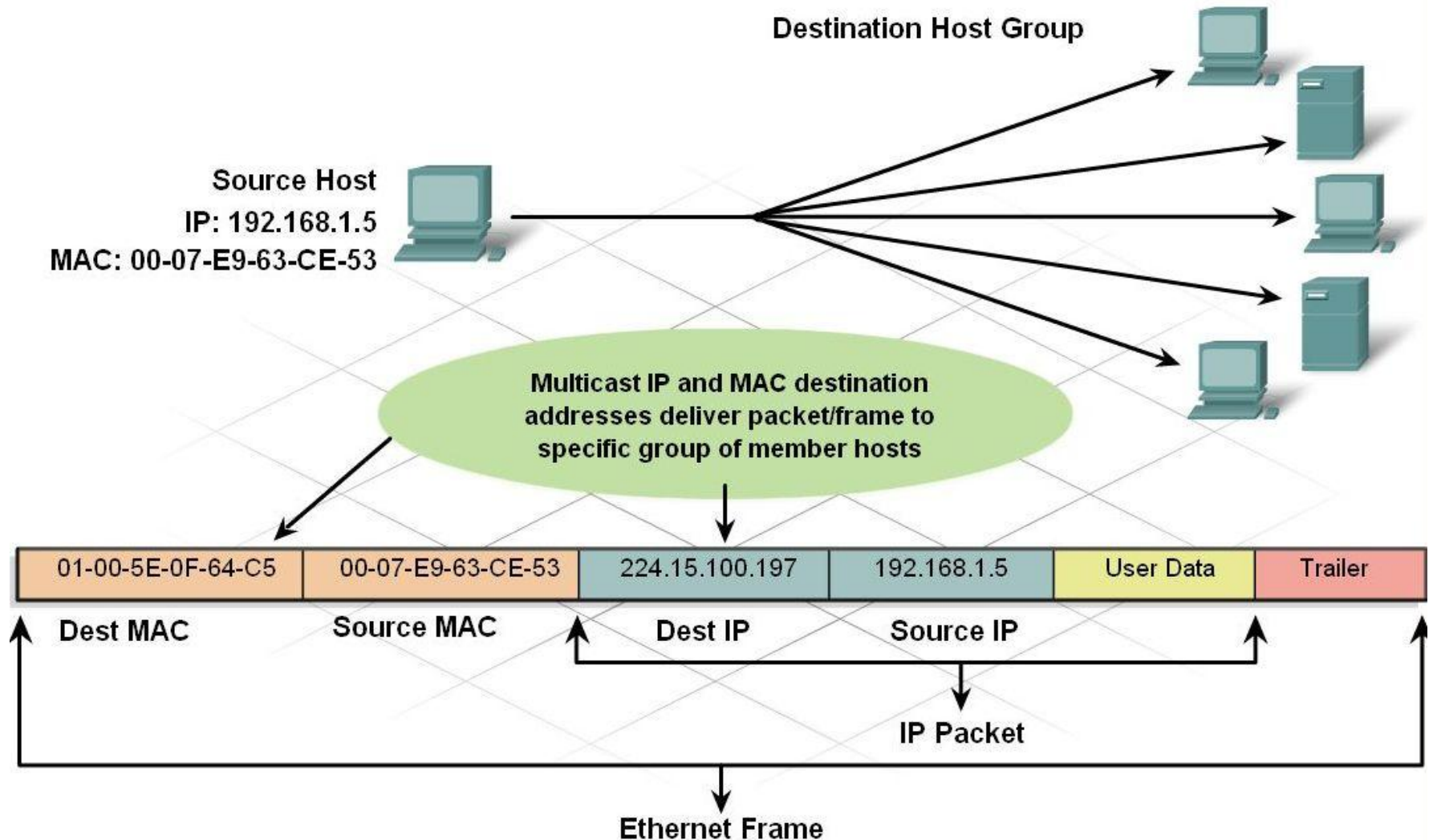
Unicasting



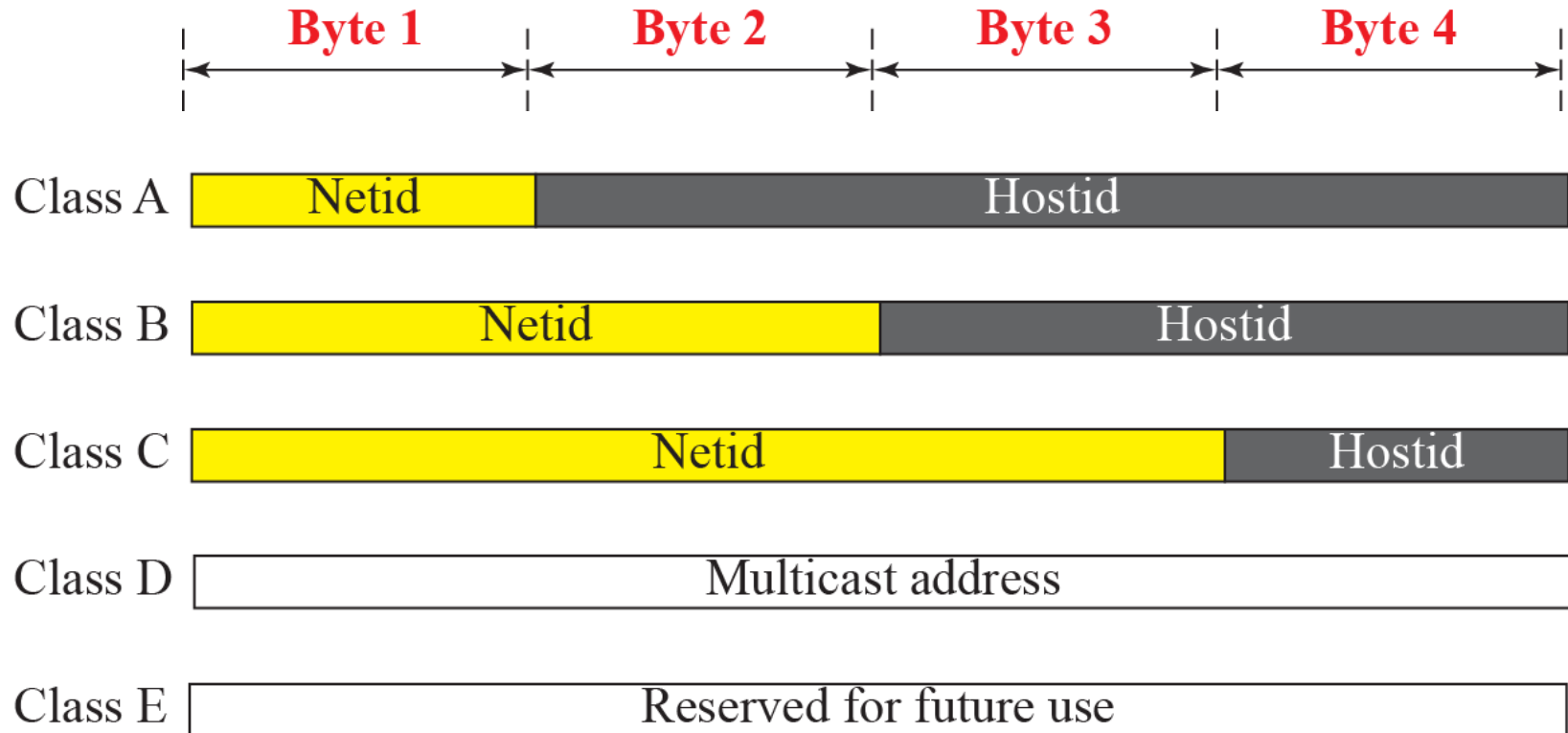
Broadcasting



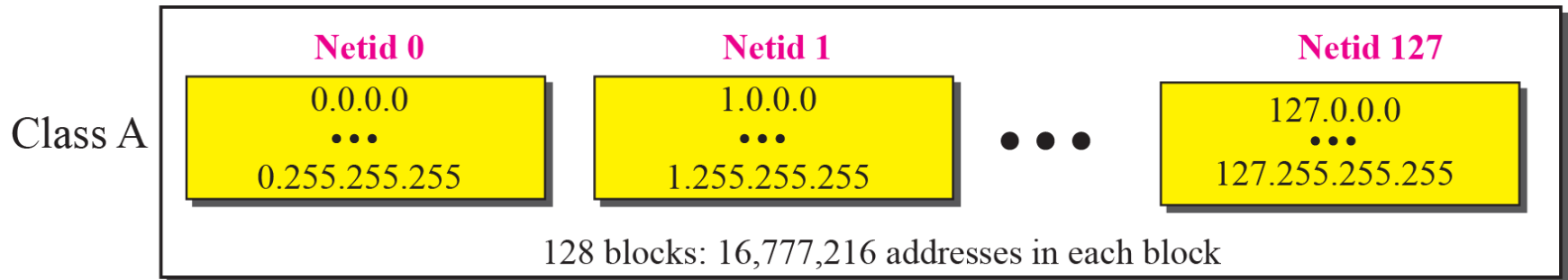
Multicasting



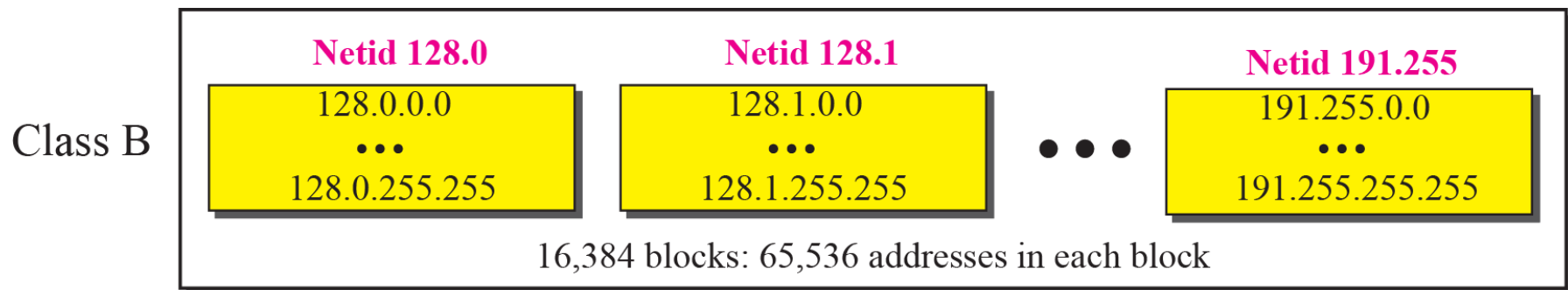
Netid and Hostid



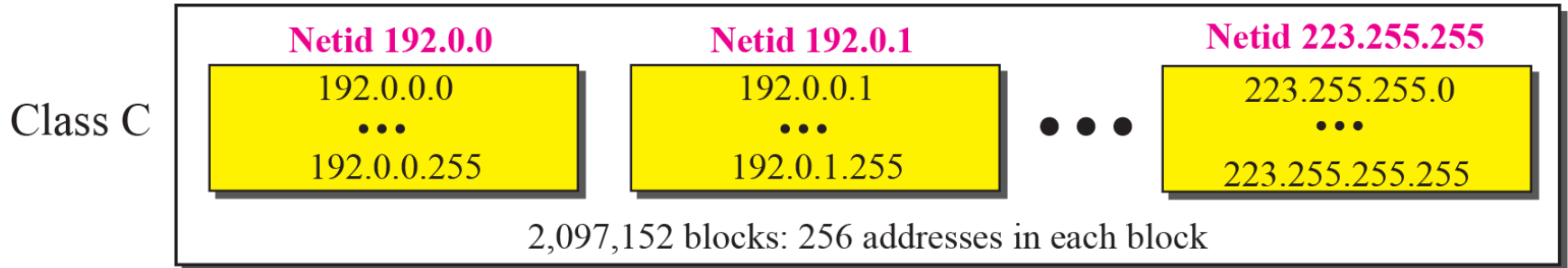
Blocks in Class A



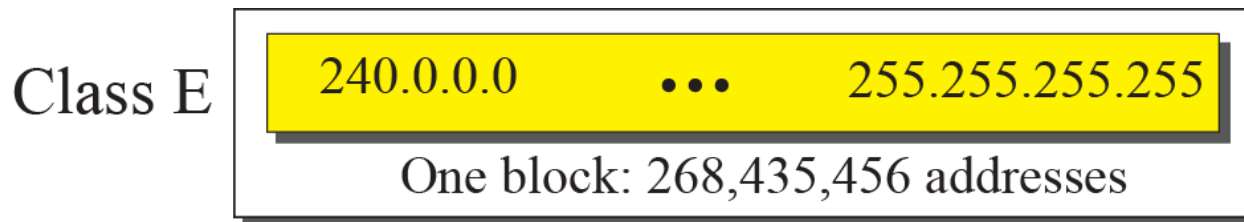
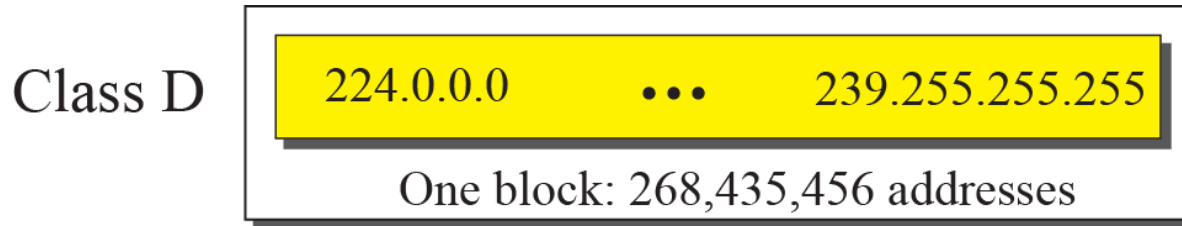
Blocks in Class B



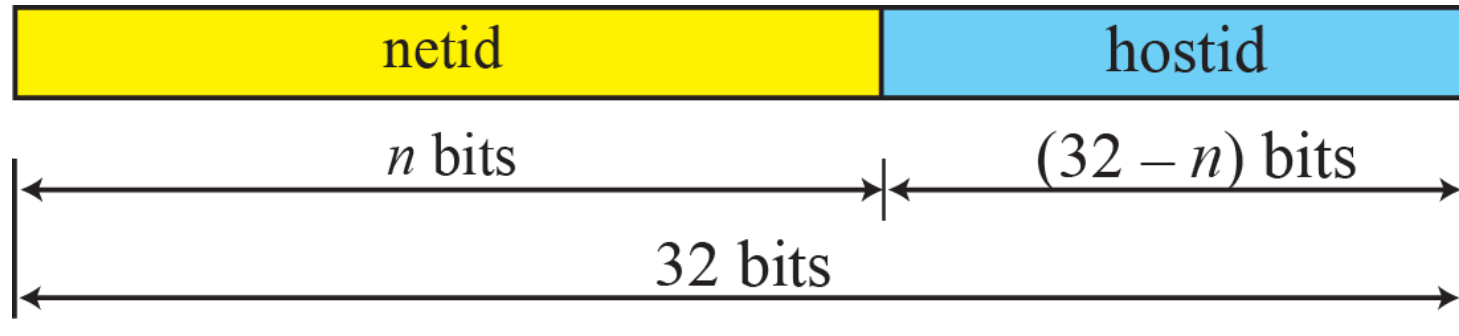
Blocks in Class C



The single block in Class D and E



Two-level addressing in classful addressing



Class A: $n = 8$

Class B: $n = 16$

Class C: $n = 24$

Example

An address in a block is given as 73.22.17.25. Find the number of addresses in the block, the first address, and the last address.

Solution

Figure shows a possible configuration of the network that uses this block.

1. The number of addresses in this block is $N = 2^{32-n} = 16,777,216$.
2. To find the first address, we keep the leftmost 8 bits and set the rightmost 24 bits all to 0s. The first address is 73.0.0.0/8, in which 8 is the value of n .
3. To find the last address, we keep the leftmost 8 bits and set the rightmost 24 bits all to 1s. The last address is 73.255.255.255.

Example

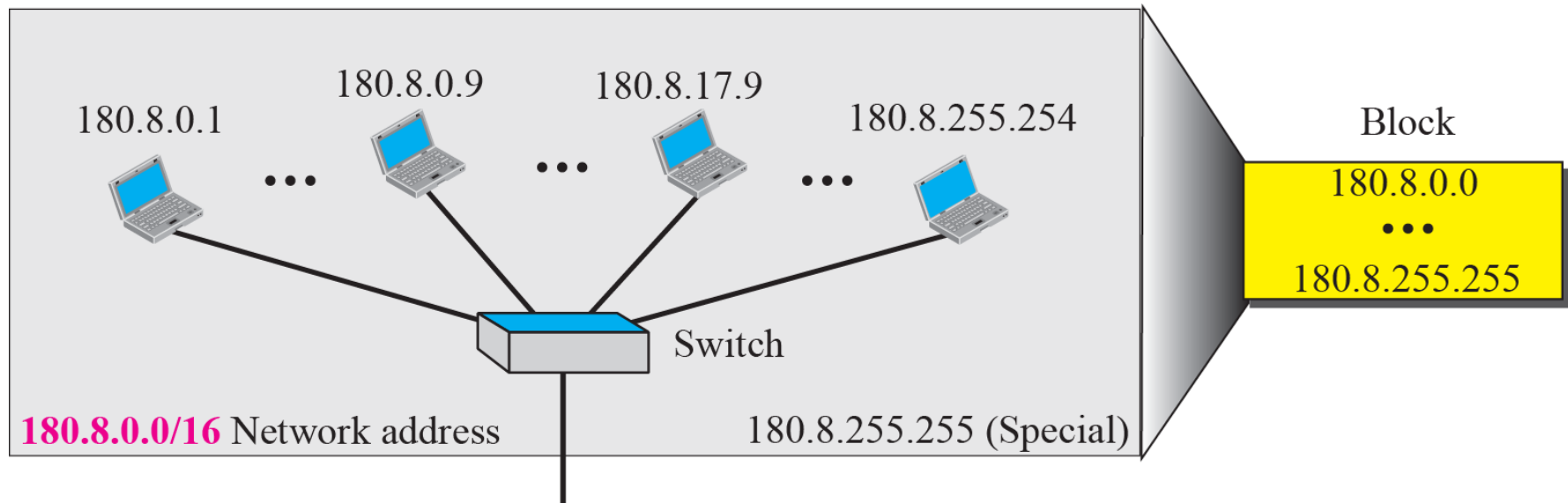
An address in a block is given as 180.8.17.9. Find the number of addresses in the block, the first address, and the last address.

Solution

Figure shows a possible configuration of the network that uses this block.

1. The number of addresses in this block is $N = 2^{32-n} = 65,536$.
2. To find the first address, we keep the leftmost 16 bits and set the rightmost 16 bits all to 0s. The first address is 180.8.0.0/16, in which 16 is the value of n .
3. To find the last address, we keep the leftmost 16 bits and set the rightmost 16 bits all to 1s. The last address is 180.8.255.255.

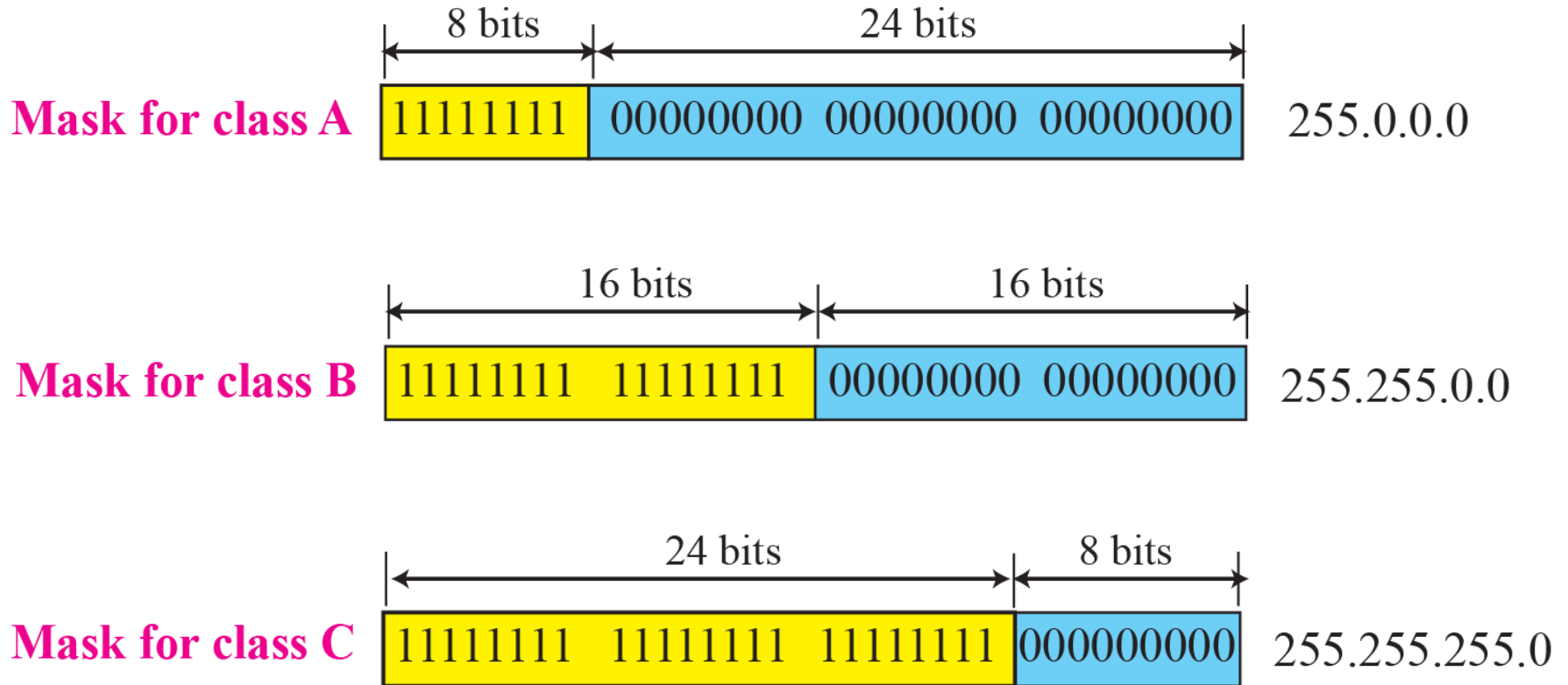
Netid 180.8: common in all addresses



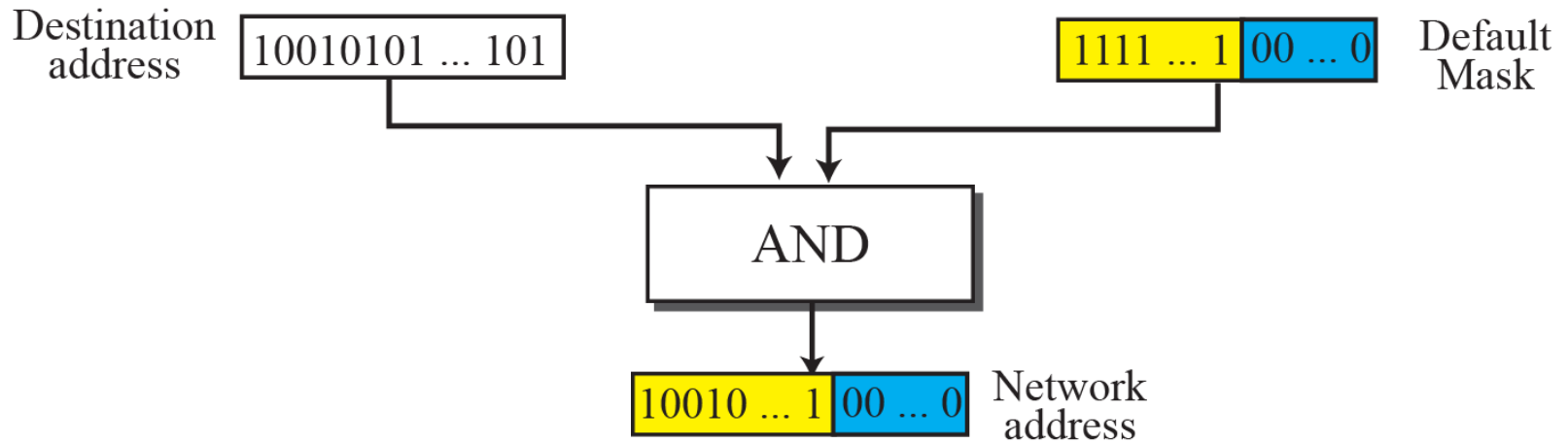
Note

The network address is the identifier of a network.

Network mask



Finding a network address using the default mask



Example

A router receives a packet with the destination address 201.24.67.32. Show how the router finds the network address of the packet.

Solution

Given: Destination IP = 201.24.67.32

Step 1: Identify the Class

The first octet is 201.

Since 192–223 belongs to Class C, this is a Class C address.

Step 2: Default Subnet Mask

Class C default mask = 255.255.255.0 (/24)

This means:

First 24 bits → Network part

Last 8 bits → Host part

- **Step 3: Find the Network Address**

To find the network address, the router performs:

- **IP Address AND Subnet Mask**
- Since the mask is **255.255.255.0**, the last octet is 0 in the mask.

This means:

- The first three octets remain the same (network part).
- The last octet (host part) becomes **0**.

So:

- 201.24.67.32
→ Replace host part (32) with 0
- **Network Address = 201.24.67.0/24**

Practice Question

A router receives a packet with the destination address 172.16.45.78.

Find the network address of the packet

Subnetting

- A **subnet (subnetwork)** is a smaller network created from a larger IP network.
- It divides one large network into multiple smaller logical networks.
- Created using **subnetting** (borrowing host bits).
- One big network is divided into multiple smaller networks called Subnets

Why Do We Need Subnets?

- Subnetting is required for:
- Reducing broadcast traffic
- Improving performance
- Better network management
- Enhancing security
- Efficient IP address usage

Need 1 – Reduce Broadcast Traffic

- In large networks, broadcast messages go to all devices.
- This increases congestion and reduces performance.
- Subnetting limits broadcast within each subnet.
- Improves speed
- Reduces unnecessary traffic

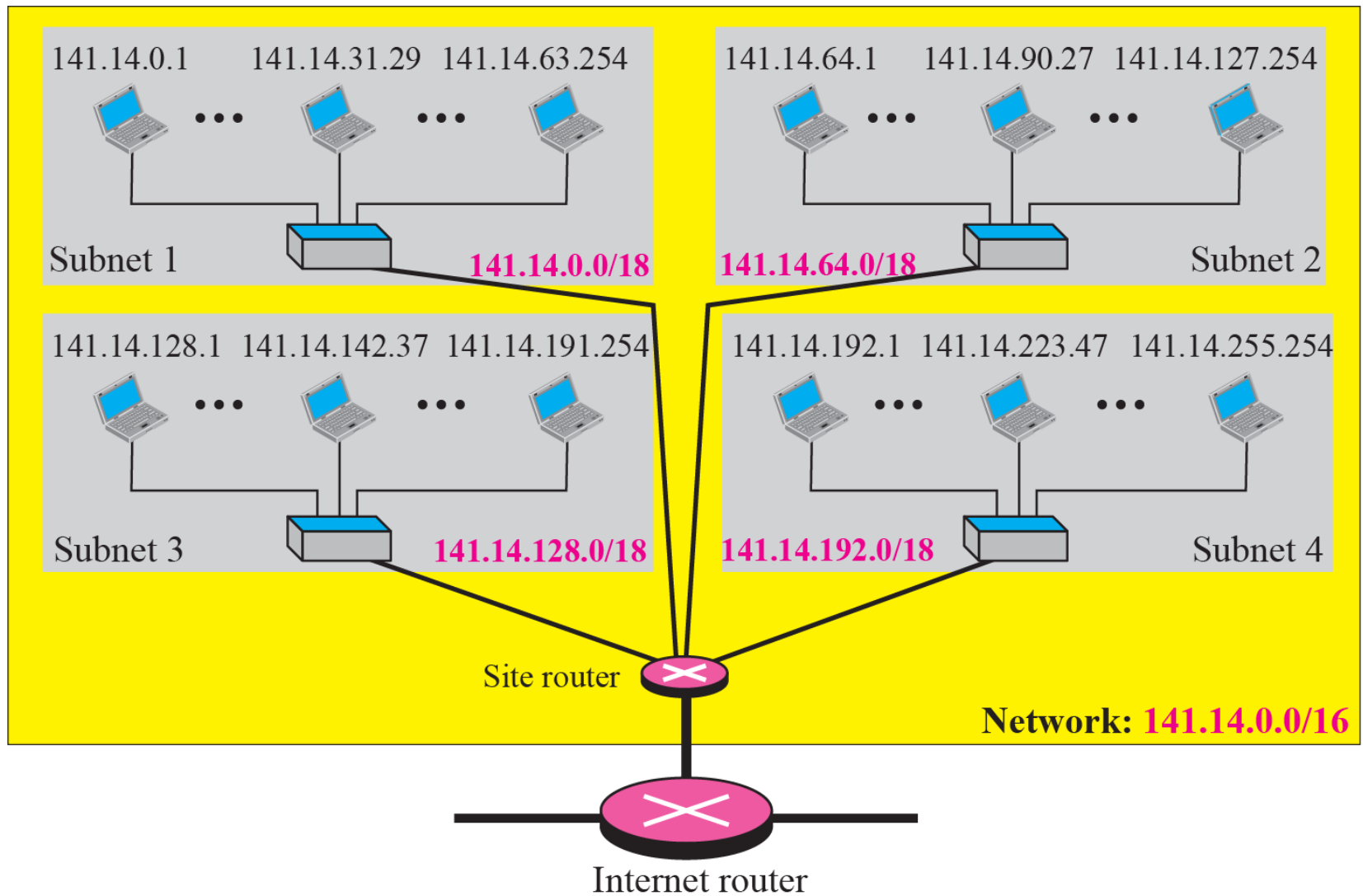
Need 2 – Better Network Management

- Different departments can use different subnets:
 - HR → Subnet 1
 - Accounts → Subnet 2
 - IT → Subnet 3
- Easier monitoring
- Organized network structure

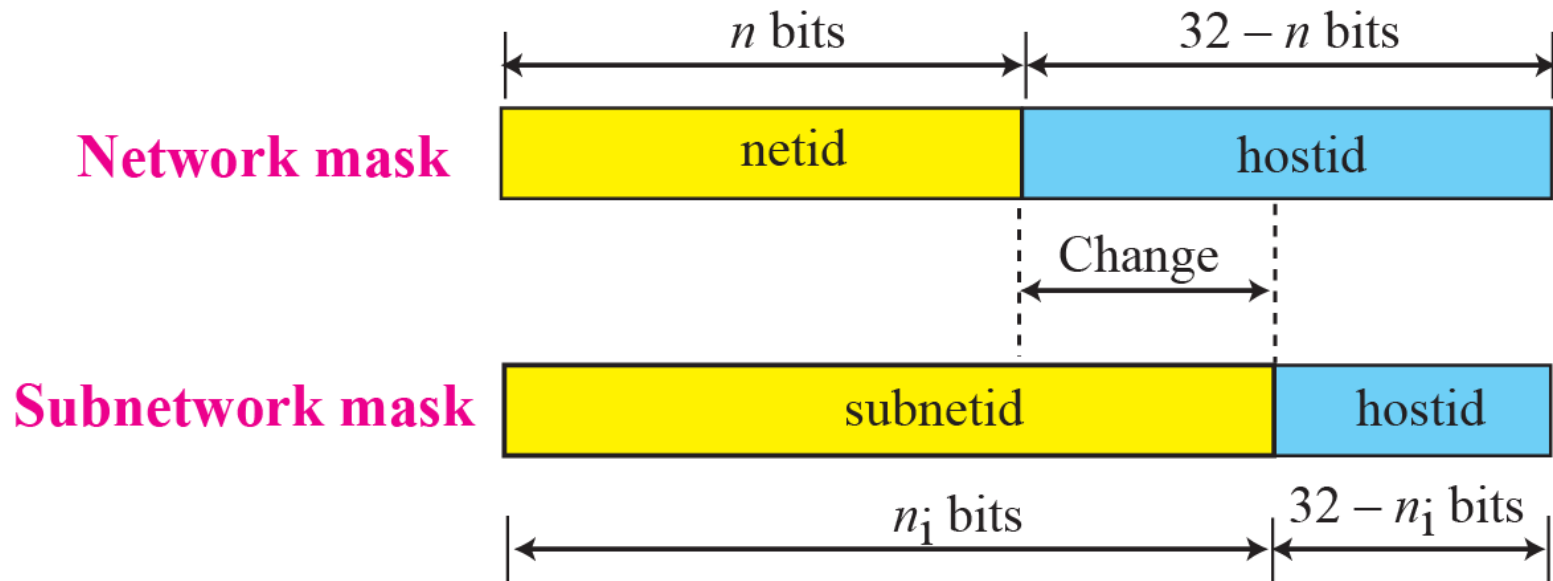
Example

Figure shows the network after subnetting.

- The whole network is still connected to the Internet through the same router.
- However, the network has used a private router to divide the network into four subnetworks.
- The rest of the Internet still sees only one network;
- internally the network is made of four subnetworks.
- Each subnetwork can now have almost 2^{14} hosts.
- The network can belong to a university campus with four different schools (buildings).
- After subnetting, each school has its own subnetworks, but still the whole campus is one network for the rest of the Internet.
- Note that /16 and /18 show the length of the netid and subnetids.



Network mask and subnetwork mask



Question 1

Given:

- Subnet mask = **255.255.255.224**

Find:

- Prefix (/ ?)
- Number of subnets (from /24)
- Hosts per subnet
- Subnet ranges for 192.168.1.0 network

Step 1: Find Prefix

- 255 → 11111111 (8 ones)
255 → 11111111 (8 ones)
255 → 11111111 (8 ones)
224 → 11100000 (3 ones)

- Total 1s:

$$8 + 8 + 8 + 3 = 27$$

- Prefix = **/27**

Step 2: Borrowed Bits

- Original Class C = /24
New = /27
- Borrowed bits:
- $27 - 24 = 3$ **bits**
- Number of subnets:
 $2^3 = 8$

Total subnets = 8

Step 3: Hosts per Subnet

- Remaining host bits:
- $32 - 27 = 5$
- $2^5 = 32$
- Usable hosts:
- $32 - 2 = 30$

Each subnet has 30 usable hosts

Step 4: Block Size

- Block size:
- $256 - 224 = 32$
- So subnet increments by 32.

Subnets of 192.168.1.0/27

Subnet	Network	Host Range	Broadcast
1	192.168.1.0	.1 – .30	.31
2	192.168.1.32	.33 – .62	.63
3	192.168.1.64	.65 – .94	.95
4	192.168.1.96	.97 – .126	.127
5	192.168.1.128	.129 – .158	.159
6	192.168.1.160	.161 – .190	.191
7	192.168.1.192	.193 – .222	.223
8	192.168.1.224	.225 – .254	.255

Question 2

Subnet the network **192.168.50.0/24** into **4 subnets**.

Find:

- New subnet mask
- Number of hosts per subnet
- Network address and broadcast address of each subnet

Step 1: Find Borrowed Bits

- We need **4 subnets**
- $2^x \geq 4$
- $2^2 = 4$
- So borrow **2 bits**

New prefix:

- $/24 + 2 =$ **$/26$**

Step 2: New Subnet Mask

- /26 means 26 ones in mask
- 255.255.255.192

New subnet mask = 255.255.255.192

Step 3: Hosts per Subnet

- Remaining host bits:
- $32 - 26 = 6$
- $2^6 = 64$

Usable hosts:

- $64 - 2 = 62$

Each subnet has 62 usable hosts

- **Step 4: Block Size**
- $256 - 192 = 64$
- Subnet increment = 64

Subnet Details

Subnet	Network Address	Broadcast Address
1	192.168.50.0	192.168.50.63
2	192.168.50.64	192.168.50.127
3	192.168.50.128	192.168.50.191
4	192.168.50.192	192.168.50.255

CLASSLESS ADDRESSING

- Subnetting and supernetting in classful addressing did not really solve the **address depletion problem**.
- With the growth of the Internet, it was clear that a larger address space was needed as a long-term solution.
- Although the long-range solution has already been devised and is called IPv6, a short-term solution was also devised to use the same address space but to change the distribution of addresses to provide a fair share to each organization.
- The short-term solution still uses IPv4 addresses, but it is called *classless addressing*.

Classless Addressing (CIDR)

- **CIDR = Classless Inter-Domain Routing**
- It is a method of IP addressing that does **not use fixed classes (A, B, C)**.
- Instead, it uses **prefix length (/n)** to define network size.

Why Classless Addressing Needed?

- Classful addressing had problems:
 - Wastage of IP addresses
 - Fixed network sizes
 - Not flexible
- CIDR solves these problems by allowing **variable-length subnet masks (VLSM)**.

Key Features of Classless Addressing

- No fixed classes
- Flexible subnet size
- Efficient IP allocation

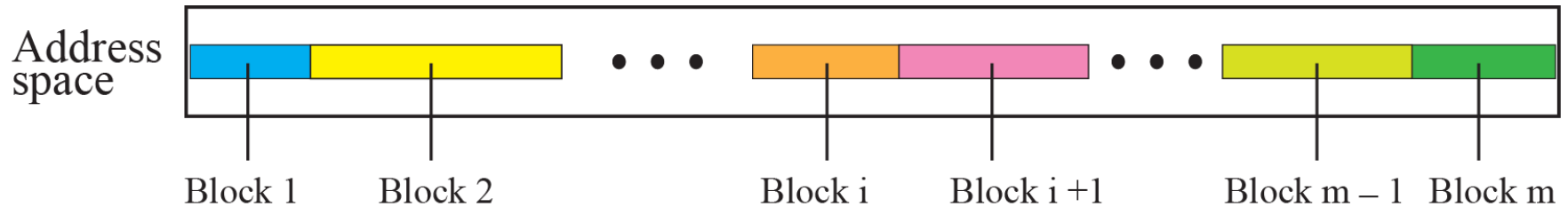
CIDR

- **Key facts**
- **Introduced:** 1993 (RFC 1519)
- **Also known as:** Supernetting
- **Core concept:** Variable-length subnet masks (VLSM)
- **Notation:** a.b.c.d/n (e.g., 192.168.0.0/24)
- **Primary benefit:** Efficient IP use and smaller routing tables

How CIDR works

- CIDR expresses an IP network as an address followed by a slash and a prefix length that indicates how many bits form the network portion.
- For instance, 192.168.1.0/24 defines a block of 256 IPs. Using variable-length subnet masks (VLSM),
- CIDR divides or combines these blocks flexibly. It also supports *supernetting*, aggregating contiguous networks into a single routing entry to streamline global routing tables

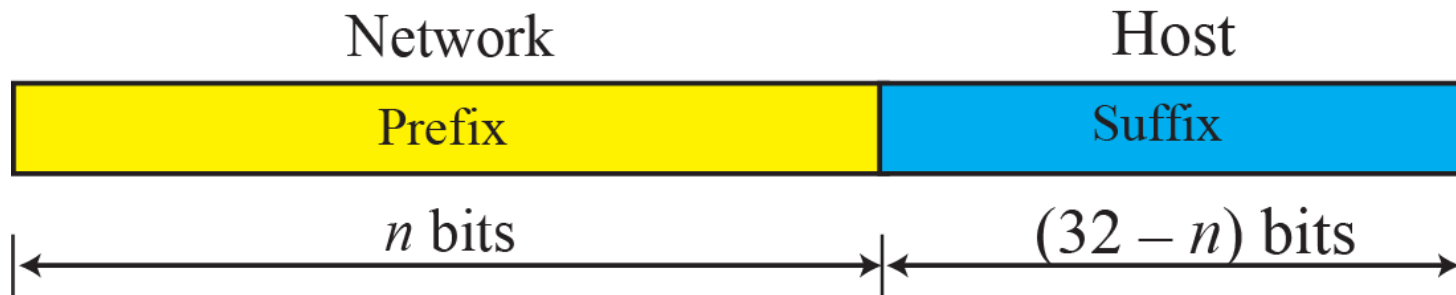
Variable-length blocks in classless addressing



Note

In classless addressing, the prefix defines the network and the suffix defines the host.

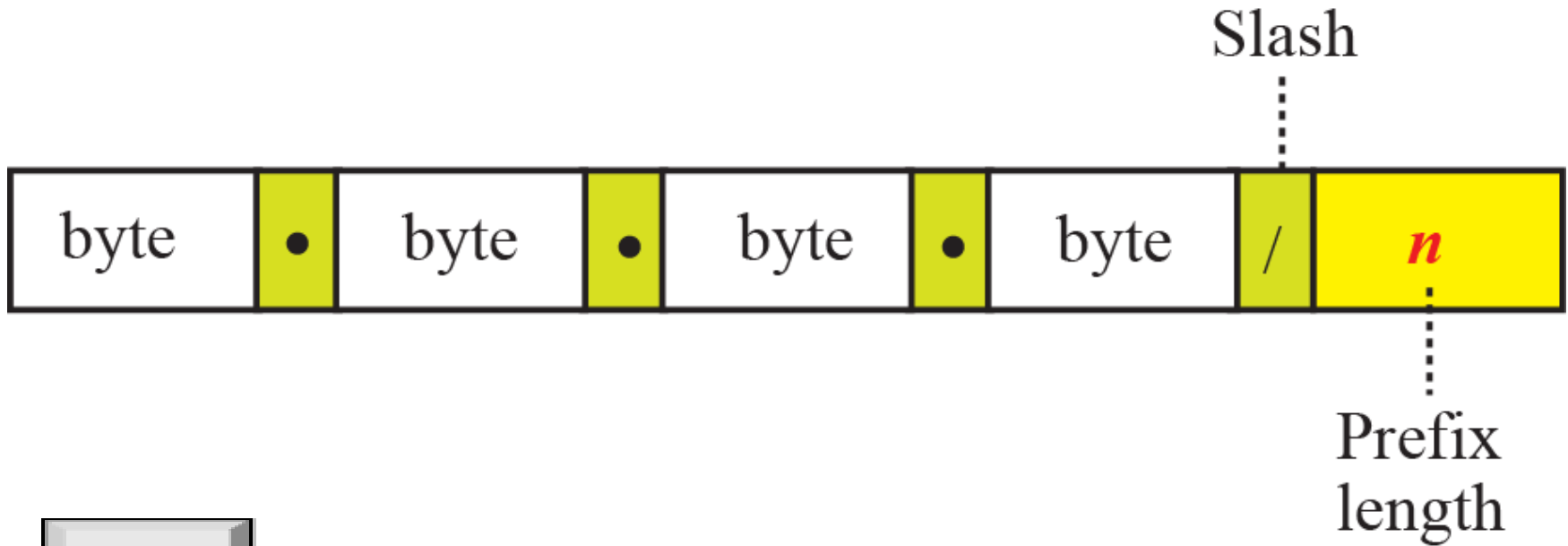
Prefix and suffix



Note

The prefix length in classless addressing can be 1 to 32.

Slash notation



Note

In classless addressing, we need to know one of the addresses in the block and the prefix length to define the block.

Question 1: CIDR

An organization is assigned the CIDR block:

150.25.32.0/21

Find:

- Total number of addresses
- Number of usable host addresses
- First address
- Last address

Step 1: Find Host Bits

- Prefix = /21
- Total bits in IPv4 = 32

Host bits:

$$32 - 21 = 11$$

Step 2: Total Number of Addresses

- $2^{11} = 2048$
- **Total addresses = 2048**

Step 3: Usable Hosts

- $2048 - 2 = 2046$
- (Network + Broadcast reserved)

Usable hosts = 2046

Step 4: Find Block Size

- /21 subnet mask: 255.255.248.0
- Block size: $256 - 248 = 8$
- So third octet increases by 8.

Given network starts at:

- 150.25.32.0

Next block would start at:

- 150.25.40.0

So range is:

- 150.25.32.0

to

150.25.39.255

- **Last address = 150.25.39.255**

Supernetting

- **Supernetting** is the process of combining multiple smaller, contiguous IP networks into one larger network.
- It is also called **Route Aggregation** or **Route Summarization**.
- Supernetting is supported through **Classless Inter-Domain Routing (CIDR)**.
- Supernetting is the process of aggregating multiple contiguous IP networks into a single larger network using CIDR notation to reduce routing complexity.

Why Supernetting is Needed?

- Reduces routing table size
- Improves routing efficiency
- Minimizes memory usage in routers
- Used by ISPs to allocate large address blocks

Thus,

- Subnetting → Divides one network into smaller networks
- Supernetting → Combines multiple networks into one larger network
- Supernetting works only with **contiguous networks**.

Example

Given four Class C networks:

- 192.168.0.0/24
- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

These can be combined into:

- **192.168.0.0/22**

How /22 is Calculated

- Each network = /24
 - Number of networks combined = 4
- $$4 = 2^2$$
- Borrowed 2 bits from network portion:
 - $24 - 2 = \mathbf{22}$
 - New Subnet Mask = 255.255.252.0

Address Range of Supernet

- Network Address → 192.168.0.0
Broadcast Address → 192.168.3.255
- Total Addresses = $2^{10} = 1024$

Total Addresses

- Host bits = $32 - 22 = 10$
- Total addresses = $2^{10} = 1024$
- Valid hosts = $1024 - 2 = 1022$

- Supernet Address → **200.10.0.0/22**
- Subnet Mask → **255.255.252.0**
- Range → **200.10.0.0 – 200.10.3.255**
- Total Valid Hosts → **1022**

Conditions for Supernetting

- Networks must be contiguous
- Number of networks must be power of 2 (2, 4, 8, 16...)
- First network address must be divisible by the total block size

Subnetting	Supernetting
Dividing one large network into smaller sub-networks	Combining multiple smaller networks into one larger network
One → Many	Many → One
Prefix length increases (/24 → /26)	Prefix length decreases (/24 → /22)
Network bits increase	Network bits decrease
Host bits decrease	Host bits increase
Fewer hosts per subnet	More hosts in combined network
Used for internal LAN design and management	Used for route aggregation and ISP routing
May increase routing table entries	Reduces routing table entries
Can be applied to a single network	Requires contiguous networks (power of 2 blocks)
Supported by CIDR	Supported by Classless Inter-Domain Routing (CIDR)

Subnetting and Supernetting

Class C Default Subnet Mask (Classful)

255 . 255 . 255	0	(Decimal)
1111 1111 . 1111 1111 . 1111 1111	0000 0000	(Binary)
NETWORK ID	HOST ID	

Class C Subnetted Network (Classless)

255 . 255 . 255	128	(Decimal)
1111 1111 . 1111 1111 . 1111 1111	1 000 0000	(Binary)
NETWORK ID	SUBNET ID	HOST ID

(Default Subnet mask with extra bits taken from Host ID)

Class C Supernet Network

255 . 255 . 254	0	(Decimal)
1111 1111 . 1111 1111 . 1111 111	0 . 0000 0000	(Binary)
NETWORK ID	SUPERNET ID	HOST ID

As you can see, with **Subnetting** we take Bits from the **Host ID** and create more **Networks** with fewer hosts, whereas with **Supernetting** we take the Bits from the **Network ID** thus giving more bits to the **Host ID** portion, which means fewer networks but more hosts per network.

Classful Addressing	Classless Addressing
IP addresses are divided into fixed classes (A, B, C, D, E)	No fixed classes
Default subnet mask is fixed (based on class)	Subnet mask is variable
Does not support VLSM	Supports VLSM (Variable Length Subnet Mask)
Less efficient IP utilization	Efficient IP utilization
Higher IP address wastage	Minimal IP wastage
Routing updates do not carry subnet mask	Routing updates include subnet mask
Example: 192.168.1.0 → automatically /24	Example: 192.168.1.0/26, /28, /30 etc.
Older addressing method	

Example 5.27

One of the addresses in a block is 167.199.170.82/27. Find the number of addresses in the network, the first address, and the last address.

Solution

The value of n is 27. The network mask has twenty-seven 1s and five 0s. It is 255.255.255.240.

a. The number of addresses in the network is $2^{32 - n} = 32$.

b. We use the AND operation to find the first address (network address). The first address is 167.199.170.64/27.

Address in binary:	10100111	11000111	10101010	01010010
Network mask:	11111111	11111111	11111111	11100000
First address:	10100111	11000111	10101010	01000000

Example *Continued*

c. To find the last address, we first find the complement of the network mask and then OR it with the given address: The last address is 167.199.170.95/27.

Address in binary:	10100111	11000111	10101010	01010010
Complement of network mask:	00000000	00000000	00000000	00011111
Last address:	10100111	11000111	10101010	01011111

Example

One of the addresses in a block is 110.23.120.14/20. Find the number of addresses, the first address, and the last address in the block.

Solution

The network mask is 255.255.240.0.

- a. The number of addresses in the network is $2^{32 - 20} = 4096$.
- b. To find the first address, we apply the first short cut to bytes 1, 2, and 4 and the second short cut to byte 3. The first address is 110.23.112.0/20.

Address:	110	.	23	.	120	.	14
Network mask:	255	.	255	.	240	.	0
First address (AND):	110	.	23	.	112	.	0

Example *Continued*

- c. To find the last address, we apply the first short cut to bytes 1, 2, and 4 and the second short cut to byte 3. The OR operation is applied to the complement of the mask. The last address is 110.23.127.255/20.

Address:	110	.	23	.	120	.	14
Network mask:	0	.	0	.	15	.	255
Last address (OR):	110	.	23	.	127	.	255

Example

An ISP has requested a block of 1000 addresses. The following block is granted.

- a. Since 1000 is not a power of 2, 1024 addresses are granted ($1024 = 2^{10}$).
- b. The prefix length for the block is calculated as $n = 32 - \log_2 1024 = 22$.
- c. The beginning address is chosen as 18.14.12.0 (which is divisible by 1024).

The granted block is 18.14.12.0/22. The first address is 18.14.12.0/22 and the last address is 18.14.15.255/22.

Example

An organization is granted the block 130.34.12.64/26. The organization needs four subnetworks, each with an equal number of hosts. Design the subnetworks and find the information about each network.

Solution

The number of addresses for the whole network can be found as $N = 2^{32} - 26 = 64$. The first address in the network is 130.34.12.64/26 and the last address is 130.34.12.127/26. We now design the subnetworks:

1. We grant 16 addresses for each subnetwork to meet the first requirement (64/16 is a power of 2).
2. The subnetwork mask for each subnetwork is:

$$n_1 = n_2 = n_3 = n_4 = n + \log_2 (N/N_i) = 26 + \log_2 4 = 28$$

Example *Continued*

3. We grant 16 addresses to each subnet starting from the first available address. Figure 5.30 shows the subblock for each subnet. Note that the starting address in each subnetwork is divisible by the number of addresses in that subnetwork.

Figure *Solution to Example*

